Course: Calculus II **Code:** MA1220 **Year:** 2025

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Calculus II (MA1220) — Problem Sheet 2

Q.1 Evaluate the limit as a definite integral by first recognizing the sum as a Riemann sum:

(i)
$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i^4}{n^5}.$$

(ii)
$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{1+\left(\frac{i}{n}\right)^2}.$$

(iii)
$$\lim_{n\to\infty} \frac{1}{n} \left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \cdots + \sqrt{\frac{n}{n}} \right).$$

Q.2 Answer the following:

(i) Let $f\colon [0,1] \to \mathbb{R}$ be defined by
$$f(x) = \begin{cases} 1 & \text{if } x \text{ is irrational,} \\ 0 & \text{if } x \text{ is rational.} \end{cases}$$
Show that f is not Riemann integrable on $[0,1]$.

(ii) Let $f\colon [0,1] \to \mathbb{R}$ be defined by
$$f(x) = \begin{cases} \dfrac{1}{x} & \text{if } 0 < x \leq 1, \\ 0 & \text{if } x = 0. \end{cases}$$
Show that f is not Riemann integrable on $[0,1]$.

(iii) Give an example of a Riemann integrable function on $[a,b]$ such that $f(x) \geq 0$ for all $x \in [a,b]$ and $\displaystyle\int_a^b f(x)\,dx = 0$, but $f(x) \neq 0$ for some $x \in [a,b]$.

Q.3 Sketch the region enclosed by the given curves and find its area:

(i) $y = |x|$ and $y = x^2 - 2$.

(ii) $y = \cos x$ and $y = 1 - \cos x$, where $0 \leq x \leq \pi$.

(iii) $x = y^2 - 4y$ and $4x + y^2 = 12$.

(iv) Sketch the region in the xy-plane defined by the inequalities $x - y^2 \geq 0$ and $1 - x - |y| \geq 0$, and find its area.

(v) Find the number a such that the line $x = a$ bisects the area under the curve $y = \dfrac{1}{x^2}$ on the interval $1 \le x \le 4$.

Q.4 Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line:

(i) $y = \ln x,\ y = 1,\ y = 2,\ x = 0$; about the y-axis.

(ii) $y = \dfrac{1}{4}x^2,\ y = 5 - x^2$; about the x-axis.

(iii) $y = x^3,\ y = 0,\ x = 1$; about $x = 2$.

Q.5 Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the specified axis:

(i) $y = 4x - x^2,\ y = x$; about the y-axis.

(ii) $y = \sqrt{x},\ x = 0,\ y = 2$; about the x-axis.

(iii) $y = x^4,\ y = 0,\ x = 1$; about $x = 2$.

(iv) $y = x^2,\ y = 2 - x^2$; about $x = 1$.

Q.6 Answer the following:

(i) Find the average value of the function $f(x) = 4x - x^2$ on the interval $[0, 4]$.

(ii) Find the average value of the function $f(x) = (x - 3)^2$ on the interval $[2, 5]$. Find c such that $f_{\text{ave}} = f(c)$.

(iii) Find the derivative of the function

$$g(x) = \int_{1-2x}^{1+2x} \sin t \, dt.$$

(iv) If

$$f(x) = \int_0^{\sin x} \sqrt{1 + t^2}\, dt, \qquad g(y) = \int_3^y f(x)\, dx,$$

find $g''\left(\dfrac{\pi}{6}\right)$.

Q.7 Determine whether each integral is convergent or divergent. Evaluate those that are convergent:

(i) $\displaystyle\int_{-\infty}^{\infty} xe^{-x^2}\, dx.$

(ii) $\displaystyle\int_0^{\infty} \sin^2 x\, dx.$

(iii) $\displaystyle\int_{-2}^{3} \frac{1}{x^4}\, dx.$

(iv) $\displaystyle\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx.$

Q.8 Answer the following:

(i) Find the length of the arc of the curve $y = 3x + 1$ for $0 \leq x \leq 3$.

(ii) Find the length of the arc of the curve $y = \ln(\cos x)$ for $0 \leq x \leq \dfrac{\pi}{4}$.

(iii) Find the area of the surface obtained by rotating the curve $x = 1 + 2y^2$ for $1 \leq y \leq 2$ about the x-axis.

(iv) The curve $y = 1 - x^2$ for $1 \leq x \leq 1$ is rotated about the y-axis. Find the area of the resulting surface.